

Gas	C_p (kJ/kg·K)	C_v (kJ/kg·K)	R (kJ/kg·K)	Molar Mass (kg/kmol)
Air	1.005	0.718	0.287	28.97
Nitrogen (N_2)	1.039	0.743	0.2968	28.013
Oxygen (O_2)	0.918	0.658	0.2598	31.999
Hydrogen (H_2)	14.307	10.183	4.124	2.016
Helium (He)	5.193	3.115	2.078	4.003
Carbon Dioxide (CO_2)	0.846	0.657	0.189	44.01
Carbon Monoxide (CO)	1.040	0.742	0.297	28.01
Methane (CH_4)	2.253	1.716	0.518	16.04
Water Vapour (H_2O)	1.864	1.403	0.4615	18.015

Table 1: Thermodynamic properties of commonly encountered ideal gases